GGG-E-2814
MARCH 17, 1993

FEDERAL SPECIFICATION

ETCHERS, ELECTRIC, PORTABLE

This Federal specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric arc and vibrating point electric etchers.

1.1.1 Federal specification coverage: Federal specifications do not cover all varieties of the commodity indicated by the title of this specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Electric etchers furnished under this specification (see 6.2) are of the following types:

Type I - Light duty electric arc etcher (500 watts)
Type II - Heavy duty electric arc etcher (1,000 watts)
Type III - Heavy duty vibrating point etcher

## 2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Preservation and Packing of Hand Tools, Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

J-C-580 - Cord, Electrical, and Wire, Electrical (0- To 600- Volt Service)

FSC 5130

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(Single copies of this specification and other Federal Specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Center in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angles, CA and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-V-173 - Varnish, Moisture and Fungus Resistant
(For the Treatment of Communication, Electronic, and Associated Electrical Equipment)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply:



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National Institute of Standards and Technology:

Handbook H-28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Underwriters' Laboratories, Inc.:

UL 73 - Motor-Operated Appliances
UL 499 - Electric Heating Appliances
UL 506 - Specialty Transformers

(Application for copies should be addressed to the Underwriters, Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Road, Northbrook IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Electrical Manufacturers Association (NEMA):

No. WC 3 - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, suite 300, Washington, DC 20037.)



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## 3. REQUIREMENTS

3.1 Illustrations. The illustrations in this specification are for the convenience of identification and are not intended to preclude the purchase of etchers that otherwise conform to this specification.

3.2 Standard product. The etcher shall be the manufacturer's standard commercial product, except for those changes necessary to comply with the specification requirements. All like items furnished, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Conformance to standards of Underwriters Laboratories, Inc. The etcher shall comply with the requirements of the specified Underwriters' Laboratories (UL) standards. The contractor shall submit proof to the procuring activity that the etcher complies to the requirements of the applicable UL standards. The UL label containing the UL listing mark will be accepted as proof of compliance to these requirements. In lieu of the UL label, the contractor shall submit independent proof, satisfactory to the Government, that the etcher complies to the requirements of the applicable UL standards including the UL methods of test. Contractor compliance with the UL requirements does not absolve the contractor from complete compliance with the requirements of this specification.

3.4 First article inspection. When specified (see 6.2), the contractor shall furnish one sample of Type I, one sample of Type II, and two samples of Type III etcher(s) for First Article Inspection. Each sample shall have been manufactured and inspected using the same materials, processes, and methods that will be used in the production of etchers to fulfill the terms of the contract or purchase order. Each sample shall meet all the requirements of this specification. Approval of the first article sample(s) shall not relieve the contractor of the responsibility to furnish etchers in accordance with the requirements of this specification.



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3.5 Materials and components. Materials and components used in the construction of the etcher shall be as specified herein. When materials and components are not specified, the materials and components shall be of the same quality used in the manufacturer's standard commercial product for the applicable purpose. If the manufacturer has more than one quality of materials and components used in the standard commercial product, the highest quality shall be used in the construction of the etcher provided under this specification.

3.6 Design, safety, and construction. Type I and II etchers shall be designed to etch on metal surfaces by the electric-arc process and shall be similar to figures 1 and 2. Type III etchers shall be designed to etch on surfaces (e.g. metal, glass, plastic, wood) by the vibratory-scarifying process and shall be similar to figure 3. Etching points shall be accessible for adjustment and replacement. The etcher shall be designed and constructed to preclude any safety hazard to the operator or environment including safety hazards caused by malfunctions of the etcher, regardless of cause.

3.7 Electrical characteristics.

3.7.1 Type I and II etchers. Each etcher shall operate on a rated voltage of 120 volts RMS at a rated current of less than 12 amperes at a rated frequency of 60 Hertz. In addition, each etcher shall be capable of operation over a voltage of 108 to 132 volts RMS at a current of less than 15 amperes over a frequency range of 50 to 60 Hertz.

3.7.2 Type III etcher. The etcher shall operate on a rated voltage of 120 volts RMS at a rated current of less than 1.0 ampere at a rated frequency of 60 Hertz. In addition, the etcher shall be capable of operation over a voltage of 108 to 132 volts RMS at a current of less than 2 amperes over a frequency range of 50 to 60 Hertz.

3.8 Insulation resistance. When tested as specified in 4.5.2, the etcher shall have an insulation resistance of 2 megohms minimum under the application for one minute of a voltage of 1000 RMS volts plus twice the rated voltage of the etcher at a frequency of 60 Hertz. The insulation resistance shall exist between all external surfaces accessible to the operator and the etcher's electrical power. In addition, the insulation resistance shall exist as specified herein.



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3.9 Power cable assembly. The power cable assembly shall be constructed with a cord conforming to Federal Specification J-C-580 (as specified below) and a three-prong grounding plug connector having the same current and voltage ratings of the cord. The plug connector shall be nonlocking and either integrally molded or attached by a cord grip to the cable jacket. The cord shall be at least 5 feet long and shall have a suitable strain relief where it passes through the housing. Cords shall be one of the following types:

| Type I and II etchers | Type III etchers |
|---|---|
| a. J-C-580-ST-3CF*/**TTJ | a. J-C-580-SJT-3CF*/**TTJ |
| b. J-C-580-SO-3CF*/**RNJ | b. J-C-580-SJO-3CF*/**RNJ |
| c. J-C-580-SJT-3CF*/**TTJ | c. J-C-580-SVT-3CF*/**TTJ |
| d. J-C-580-SJO-3CF*/**RNJ | |

NOTE: * Two-conductor cable with additional grounding conductor
** 18 AWG for types I and III. 16 AWG for type II

3.10 Fasteners. Screws, pins, bolts, and similar fasteners shall be installed with means of preventing loss of tightness or separation. Fasteners and parts subject to removal or adjustment, shall not be swaged, preened, staked, or permanently deformed. Threaded fasteners and parts shall be in the inch system and conform to Handbook H-28.

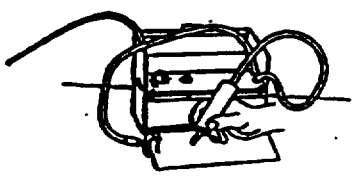


FIGURE 1. Type I

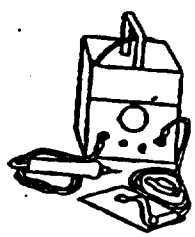


FIGURE 2. Type II

3.11 Type I and II etcher. Each etcher shall conform to the requirements of UL 499. Each etcher shall produce permanent surface discoloration of a metallic work piece due to oxidation induced by high temperature at the point of electrical contact. Each etcher shall have components consisting of a case, transformer, control panel, power "ON-OFF" switch, power cable with plug, etching tool lead, ground lead, stylus, and when furnished, a ground plate.

3.11.1 Type I heat output. The heat output for a Type I etcher shall be a minimum of 500 watts with 4 heat-taps. Three of the heat-taps shall be below 300 watts. When tested as specified in 4.5.1.2, the heat-taps shall vary in output from a minimum of 150 watts to the maximum heat output of the etcher.



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3.11.2 Type II heat output. The heat output for a Type II etcher shall be a minimum of 1000 watts with 10 heat-taps. At least three of the heat-taps shall be below 300 watts. When tested as specified in 4.5.1.4, the heat-taps shall vary in output from a minimum of 150 watts to the maximum heat output of the etcher.

3.11.3 Weight. The weight of a Type I etcher shall not exceed 25 pounds. The weight of a Type II etcher shall not exceed 35 pounds.

3.11.4 Components.

3.11.4.1 Case. The case shall be made of sheet steel of 20 gage (.0359 inch) minimum thickness. The base of the case shall have five embossed dimples of 3/8-inch spherical radius (inside) that shall raise the base of the etcher case. One dimple shall be located at the geometrical center and the remaining four dimples shall be located 3/4-inch from each side of each corner of the base. Alternatively, rubber pads of 7/8-inch diameter by 3/8-inch thick minimum dimensions may be substituted for the dimples. The pads shall be attached with locked screws. The case shall have a handle with a hand grip. Space shall be provided within the case for stowage of the power cable, stylus with spare etching points, stylus and ground leads, and (when furnished by the contractor) the ground plate. Alternatively, an integral part of the case, suitable for use as a ground for the etching tool, may be substituted for the ground plate (see 3.11.4.6).

3.11.4.2 Transformer. The transformer shall conform to UL 506. The transformer shall be designed and capable of operation consistent with the etcher electrical characteristics (see 3.7) and the requirements of this specification. The transformer temperature shall not exceed 65 degrees Celsius above ambient temperature under any operating conditions. The transformer shall be securely mounted in the etcher case so that the transformer does not become loose or detached from its mounting. The use of an adhesive for mounting is not acceptable.

3.11.4.3 Control panel. The control panel shall have an "ON-OFF" power switch, a light to indicate when power is applied, two output terminal studs with insulated knurled contact nuts, and a rotary heat-tap selector with a pointer and legible metallic dial for indicating the applicable heat-taps (see 3.11.1 and 3.11.2). The heat-tap dial shall be securely attached to the panel by rivets. Other means, such as jack plug receptacles, may be provided in lieu of the rotary heat-tap switch. The output terminals shall be insulated to prevent any grounding through the hole cross sectional and panel surface areas.



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3.11.4.4 Stylus and ground leads. Both, the stylus lead and the ground lead shall be 3 feet or longer of size 4 AWG "Neoprene 600V" welding cable, or a cable of equal or better characteristics, conforming to NEMA WC 3. Asbestos-insulated wire is not acceptable. Both leads shall be equipped with a spade-type copper terminal at one end for connection to the transformer terminal studs. A stylus shall be attached to the other end of the stylus lead and a positive-type grounding copper terminal shall be attached to the other end of the ground lead.

3.11.4.5 Etching tool (stylus). The etching tool (stylus) shall have an insulated holder of suitable size and shape for ease of handling and replacement of the etching point. Unless otherwise specified (see 6.1), the etching tool (stylus) shall be provided with a spare etching point. The etching point shall be copper, steel, or copper alloy and shall have integral tips or inserts of tungsten carbide or tungsten. The etching point shall withstand high temperatures without distortion or blunting of the tip that would make etching illegible. Etching points in standard commercial use of equal or better materials and functional performance as the points specified are acceptable.

3.11.4.6 Ground plate. When furnished by the contractor, the ground plate shall have a minimum of 25 square inches of surface area. The plate shall be a minimum of 18 gage (.0478 inch) steel and shall be copper plated for better electrical conductivity. The ground plate may be furnished as an integral part of the case.

3.11.5 Water and fungus protection. When specified (see 6.2), the etcher's electrical circuitry (including all components and connections) shall be protected from the affects of water and fungus growth except as follows:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be protected.

(b) Components or circuit elements whose function would be adversely affected by the protection shall not be protected.

3.11.5.1 Varnish. Protection from water and fungus growth shall be provided by treatment with a varnish conforming to MIL-V-173. The varnish shall be applied to components and surfaces previously cleaned and prepared in accordance with best commercial practice. The varnish shall be applied to provide a minimum dry film thickness of 0.001 inch.



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3.11.6 Drop requirement. Each etcher shall withstand a 3 foot drop onto a concrete surface (landing on any etcher case corner) with no breakage or deformation that renders the etcher functionally defective.

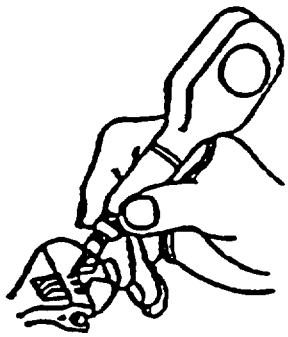


FIGURE 3. Type III

3.12 Type III. The etcher shall conform to the requirements of UL 73. The etcher shall produce permanent indentations in the work piece material (metal, glass, plastic, wood) due to the repetitive impact of the etcher point. The etcher shall be self-contained and shall be enclosed in a rugged housing. The etcher shall be similar to figure 3 and shall consist of a housing, drive unit, etching point shaft with etching point, collet or chuck, adjusting knob, power switch, and power cord with plug.

3.12.1 Weight. The etcher shall weigh 1 pound or less.

3.12.2 Components.

3.12.2.1 Housing. The housing shall be of a size and shape that forms a comfortable handgrip for an operator and shall be constructed of a material that will insulate the operator from electrical power. The housing material shall be in accordance with the insulation resistance requirements (see 3.8).

3.12.2.2 Drive unit. The drive unit shall consist of a heavy duty vibrating unit utilizing an electric coil arranged for equal distribution of magnetic flux around the armature. The drive unit and coil shall be securely held in place.



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3.12.2.3 Etching point shaft. The etching point shaft material shall be a properly hardened alloy steel. The shaft shall extend from a resting position against a vibrating armature through a collet or chuck. The etching point shaft shall be separated from the drive unit (see 3.12.2.2) in a manner which prevents the etching point shaft from becoming electrically energized should the vibrating armature become electrically energized. The shaft shall be spring loaded and shall cycle at a minimum of 7200 strokes per minute at rated electrical power (see 3.7.2). With maximum electrical power (see 3.7.2) applied to the etcher, the etching point shaft shall be capable of being locked for two minutes without causing damage to the etcher or safety or fire hazard. Unless otherwise specified (see 6.1), six replacement etching point shafts shall be furnished with each etcher. Two of the shafts shall be diamond tipped and four of the shafts shall be tungsten carbide tipped.

3.12.2.4 Collet or chuck. A collet or chuck shall be provided for replacement of the etching point shafts. The collet or chuck shall be separated from the drive unit (see 3.12.2.2) in a manner which prevents the collet or chuck from becoming electrically energized should the vibrating armature become electrically energized.

3.12.2.5 Adjusting knob. The etcher adjusting knob shall change the length of stroke and hence the depth of cut and width of the line being marked. The adjusting knob shall not be capable of being adjusted in a manner or to a setting that causes the etcher to be a safety or fire hazard. The adjusting knob shall be electrically insulated (see 3.8) in a manner that precludes any safety or fire hazard including those that might occur if the etcher should malfunction. Prior to shipment, the adjusting knob shall be adjusted to the position that provides minimum length of stroke.

3.12.2.6 Power Switch. An "ON-OFF" power switch of proper rating shall be provided in the etcher housing at a point convenient to the operator. The power switch shall be a toggle, slide, or rocker-type switch. The power switch shall be electrically insulated (see 3.8) in a manner that precludes any safety or fire hazard including those that might occur if the etcher should malfunction.



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3.13 Performance.

3.13.1 Type I etcher. The Type I etcher shall be capable of operating for 30 minutes at a heat output of 500 watts in the "ON" portion of the duty cycle. The etcher shall be capable of providing legible etching during and subsequent to the continuous etching operation.

3.13.2 Type II etcher. The Type II etcher shall be capable of operating for 30 minutes at a heat output of 1000 watts in the "ON" portion of the duty cycle. The etcher shall be capable of providing legible etching during and subsequent to the continuous etching operation.

3.13.3 Type III etcher. The Type III etcher shall be capable of continuous etching operation over 4 consecutive fifteen minute intervals with 2 minutes between intervals. The etcher shall be capable of providing legible etching during and subsequent to the continuous etching operation.

3.14 Finish. Each etcher shall have enameled or plated finishes on metal surfaces in accordance with the manufacturer's standard commercial practice. In addition, the Type I and II etcher shall have an enameled coat on the outside and inside surfaces of the case. Cadmium plating shall not be used.

3.15 Identification plate. A corrosion resistant metallic or plastic identification plate that is permanently and legibly marked with the following information shall be securely attached to the outside surface of each etcher:

Manufacturer's name
Manufacturers's model designation
Manufacturer's assigned serial number (Types I & II only)
Voltage rating, AC only
Frequency rating
A caution statement reading (or the equivalent) "CAUTION: For safe operation see instruction manual"

3.16 Accessories. In addition to the specified accessories and spares, each etcher shall be furnished with those accessories and spares that are normally provided commercially with the etcher. Additional accessories shall be as specified (see 6.1)

3.17 Workmanship. The workmanship of the etcher shall be of the highest quality consistent with best commercial practice. The finish coat shall be adherent, smooth and free of runs, tackiness, and unfinished areas. The etcher shall be free of loose parts and components, debris, burrs, slivers, and sharp edges.



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3.18 Instruction sheet and parts list. Unless otherwise specified (see 6.2), the contractor shall furnish two copies of an instruction sheet and a list of replacement parts and accessories including source(s) of supply with each etcher. The instructions shall delineate and highlight appropriate electrical power requirements (voltage, current, frequency), safety requirements (e.g. fuses, circuit breakers, grounding), recommended safety protections (e.g. safety glasses), operator safety precautions, and warnings for safe operation of the etcher. The instructions shall include all information necessary for safe and successful operation and maintenance of the etcher.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspections shall be classified as follows:

(a) First article inspection.

(b) Quality conformance inspection.

4.2.1 First article inspection. First Article Inspection shall be applied to the first article sample(s) submitted in accordance with 3.4. Failure of any first article sample to pass the examinations is cause for rejection. Unless otherwise specified (see 6.2), First Article Inspection shall consist of the examinations in 4.4.1, 4.4.1.1 and applicable tests of 4.5.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All etchers of the same type offered for delivery at one time shall be considered a lot for the purpose of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in accordance with MIL-STD-105 at Inspection Level II.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each sample selected in accordance with 4.3. Quality conformance inspection shall consist of the following:

(a) Examination in accordance with 4.4.1 and 4.4.1.1.

(b) Applicable tests in accordance with 4.5.

(c) Inspection of preparation for delivery in accordance with 4.6.



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4.4.1 Examinations. Each sample shall be examined to determine compliance with the requirements of this specification. Examinations shall be conducted in accordance with Table I with the Acceptable Quality Level (AQL) as specified. Any deviation from the requirements shall be cause for rejection. Rejected samples shall not be included in a lot for delivery and if the number of rejects exceeds the acceptance number for that sample, the lot shall be rejected.

4.4.1.1 Code and standard examinations. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards that are submitted as proof of compliance with specification requirements shall be examined and validated.



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TABLE I. Classification for defects

| Defects | AQL percent defective | | |
|---|---|---|---|
| | Critical 0.10 | Major 1.0 | Minor 2.5 |
| Etcher type incorrect | X | | |
| Compliance with UL standards not as specified | X | | |
| Safety not as specified | X | | |
| Materials, components, or assemblies not as specified | | X | |
| Design or construction not as specified | | X | |
| Fasteners not as specified | | X | |
| Finish not as specified | | X | |
| Identification plate not as specified | | X | |
| Accessories not as specified | | X | |
| Weight not as specified | | | X |
| Water and fungus growth protection not as specified | | | X |
| Workmanship not as specified | | | X |



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4.5 Tests.

4.5.1 Performance test. Unless otherwise specified, the performance tests shall be conducted at ambient temperature with rated electrical power (see 3.7) applied to the etcher.

4.5.1.1 Type I operation. The power switch shall be turned "ON" and the etcher shall be operated at 500 watts for a total etching time of 30 minutes during an elapsed time interval of not more than 180 minutes. The etcher shall etch letters and numerals on a 1/8 inch thick cold rolled steel plate. The duration of the etching time cycles shall be determined by the temperature of the transformer which shall not be allowed to increase above the ambient temperature by more than 65 degrees Celsius. Overheating, failure to produce legible markings, failure to sustain the required etching time within the specified elapsed time, any degradation or failure in operation, safety or durability of the etcher shall constitute failure to pass this test.

4.5.1.2 Type I power output. With 108 volts RMS at a frequency of 50 Hertz applied (see 3.7), the power switch shall be turned "ON" and the power at the output terminal for each heat-tap setting of the etcher shall be measured to determine compliance with the heat output requirements of 3.11.1. This test shall be repeated at 132 volts RMS at a frequency of 60 Hertz. Any non-conformance at either input voltage shall constitute failure to pass this test.

4.5.1.3 Type II operation. The power switch shall be turned "ON" and the etcher shall be operated at 1000 watts for a total etching time of 30 minutes during an elapsed time interval of not more than 180 minutes. The etcher shall etch letters and numerals on a 1/8 inch thick cold rolled steel plate. The duration of the etching time cycles shall be determined by the temperature of the transformer which shall not be allowed to increase above the ambient temperature by more than 65 degrees Celsius. Overheating, failure to produce legible markings, failure to sustain the required etching time with the specified elapsed time, any degradation or failure in operation, safety or durability of the etcher shall constitute failure to pass this test.

4.5.1.4 Type II power output. With 108 volts RMS at a frequency of 50 Hertz applied (see 3.7), the power switch shall be turned "ON" and the power at the output terminal for each heat-tap setting of the etcher shall be measured to determine compliance with the heat output requirements of 3.11.2. This test shall be repeated at 132 volts RMS at a frequency of 60 Hertz. Any non-conformance at either input voltage shall constitute failure to pass this test.



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4.5.1.5 Type III operation. The power switch shall be turned "ON" and the etcher shall operate for a total of 60 minutes in four consecutive fifteen minute intervals with 2 minutes between intervals. The etcher shall etch legible letters and numerals on a 1/8 inch thick cold rolled steel plate having a hardness of 68 to 70 on the Rockwell B scale. Overheating, failure to produce legible markings, failure to sustain the required etching within the specified elapsed time, any degradation or failure in operation, safety or durability of the etcher shall constitute failure to pass this test.

4.5.1.6 Type III low voltage-frequency operation. With 108 volts RMS at 50 Hertz applied (see 3.7), the adjustment knob adjusted for maximum length of stroke of the etching point shaft, and the power switch "ON", the etcher shall etch legible letters and numerals for 5 minutes on a 1/8 thick cold rolled steel plate having a minimum hardness of 68 to 70 on the Rockwell B scale. Failure to produce legible markings, failure to sustain the required etching over the specified time, any degradation or failure in operation, safety or durability of the etcher shall constitute failure to pass this test.

4.5.1.7 Type III etching point shaft speed. The power switch shall be turned "ON" and the speed of the etching point shaft shall be measured to determine compliance with the speed requirements of 3.12.2.3. Any non-conformance shall constitute failure to pass this test.

4.5.1.8 Type III locked etching point shaft. With 132 volts RMS at a frequency of 60 Hertz applied (see 3.7), the adjustment knob adjusted for best etching, and the power switch "ON", the etcher shall be brought to a stalled condition by pressing the etching point on a metallic surface until movement ceases between the etching point and the etcher's housing. This locked etching point shaft condition shall be maintained for two minutes. The etcher shall not create a safety or fire hazard in this condition. After two minutes, the pressure shall be removed and the etcher shall resume normal operation. Any nonconformance shall constitute failure to pass this test.

4.5.2 Insulation resistance test. After successful completion of the performance tests (see 4.5.1), each sample shall be tested for insulation resistance as follows:



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4.5.2.1 Type I and II. With the power switch turned "ON", the insulation resistance of the etcher shall be measured to determine compliance with the requirements of 3.8. The specified voltage shall be applied between the power input terminal of the applicable etcher as the positive terminal and each of the two ground terminals as follows:

(a) A power output terminal with the etcher set on the maximum heat-tap, and

(b) The etcher case.

4.5.2.2 Type III. With the power switch in the "ON" position, the insulation resistance of the etcher shall be measured to determined compliance with the requirements of 3.8. The specified voltage shall be applied between the power input terminal of the etcher as the positive terminal and the following ground terminals:

(a) Etching point collet or chuck.

(b) Etching point shaft.

(c) Power switch.

(d) Housing including all shell sections.

(e) Adjustment knob.

4.5.2.3 Conformance. Any non-conformance shall constitute failure to pass this test.

4.5.3 Drop test.

4.5.3.1 Type I and II. The etcher shall be drop tested to determine compliance with the requirements of 3.11.6.

4.5.3.2 Conformance. There shall be no breakage of the etcher. Any safety or environment hazard, any failure, or any sign of disengagement of parts or failure to resume normal operation shall constitute failure to pass this test.

4.6 Inspection of preParation for delivery. An examination shall be made to determine that preservation, packaging, packing, and marking are in accordance with PPP-P-40 as required by section 5.



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## 5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, packing, andmarking for each level of protection shall be in accordance with the applicable requirements of PPP-P-40 (see 6.2).

## 6. NOTES

6.1 Intended use.

6.1.1 Type I. This etcher is intended for etching tools and other objects of ferrous alloy material within the capacity of the etcher.

6.1.2 Type II. This etcher is intended for use in etching heavy ferrous materials including tools and heavy castings within the capacity of the etcher.

6.1.3 Type III. This etcher is intended for use in etching practically all medium hardness materials including ferrous alloy materials.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Etcher type required (see 1.2).
(c) First Article testing, if required (see 3.4).
(d) Number of samples to be submitted for First Article testing if other than specified (see 3.4).
(e) Number of replacement etching points required, if other than specified (see 3.11.4.5 and 3.12.2.3).
(f) Type of points desired, if other than specified (see 3.11.3.6 and 3.12.2.3).
(g) Water and fungus protection, when required (see 3.11.5).
(h) Accessories required (see 3.16).
(i) Instruction sheet requirement, if other than specified (see 3.18).
(j) Level of preservation and packaging and level of packing required (see 5.1).

Preparing Activity: GSA-FSS



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